

Wireless Presentation and Conferencing System

SW-640L-TX-W Marketing FAQ

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1 Conferencing System

1.1 Can Support Background BYOD Connection

Q

Once a BYOD source is stopped shown, its connection or session becomes unavailable too. Consequently, the user cannot recall this BYOD source unless he/she initiates the BYOD session from the source device once again. Can the device support the background BYOD connection so that the user can recall the BYOD source quickly?

A

The BYOD switcher does not support the background BYOD connection, the main reason is privacy issues. We can imagine a typical usage scenario where user A wireless casting his smartphone to the BYOD switcher and eventually displays it on the TV. Then user B connects his laptop to the BYOD switcher via USB-C, and the TV automatically switches to the laptop's screen content.

At this time, user A will think that his smartphone connection has been disconnected, and user A may be handling some personal matters on the smartphone. If the BYOD switcher has been maintaining a background BYOD connection for the smartphone, when user B disconnects the laptop's USB-C cable, the TV will automatically display the content of the smartphone, which may cause the private content that user A is operating on the phone to be exposed.

We believe that such a scenario is unacceptable to ordinary users. Therefore, when the BYOD source is stopped shown, its connection will also be terminated to protect user's privacy.

1.2 Why not Support USB Capture Card

Q

The USB capture card feature can connect the BYOD switcher directly to Microsoft Teams on the PC, why does the device not support USB capture card?

A

Teams have certification for USB capture cards. When a certified USB capture card is recognized by Teams, the video output by the USB capture card is considered as presentation content, not a camera.

When Teams sends presentation content to the remote end, it will be encoded with higher definition (usually greater than 720p) and lower frame rate (usually less than 10 frames per second) to ensure the clarity of presentations such as PPT documents; while for ordinary cameras, it will be encoded with lower definition and higher frame rate to ensure the smoothness of the people screen.

Therefore, if we add USB capture card hardware to the BYOD switcher, it will only be used as an ordinary camera by Microsoft and cannot meet the high-definition requirements of PPT file sharing. At the same time, Microsoft has stopped certifying new USB capture cards, so when customers have USB capture card needs, the BYOD switcher needs to work with an external certified USB capture card (the relevant list of certified USB capture cards can be found on the Teams official website).

1.3 What Work Modes Do Miracast Have?

Q

The terms like Miracast over P2P network, Miracast over infrastructure network is seen often. What work modes do Miracast have? What work modes are supported now?

A

This question involves many aspects which will be described in several chapters.

1.4 What is Infrastructure Network?

In Miracast related documents, you often encounter the term "infrastructure network". In short, it refers to a network built on network infrastructure such as Wi-Fi AP and/or network switch. Generally, there are two ways to build or connect to an infrastructure network:

1. Wired With a network cable, maybe CATx copper cable or optical fiber.
2. Wireless Through a Wi-Fi AP

Accordingly, there are two typical types of infrastructure networks:

1. Conventional wired network
2. A wireless network where all hosts are connected to one or multiple Wi-Fi AP(s)

Of course, an infrastructure network can consist of both wired and wireless connections too.

2 Miracast Working Modes

2.1 Principles

The mechanism of Miracast is relatively flexible, from the perspective of an end user, it can be roughly divided into four modes.

1. Wi-Fi Direct (P2P) mode

The source and sink are connected directly by Wi-Fi direct protocol, no need for infrastructure network. This mode is mandatory, every Miracast compliant source and sink must support this mode.



Figure1. Wi-Fi Direct (P2P) mode

2. TDLS mode

The source and sink are connected to a wireless infrastructure network already, they build wireless connection with TDLS (IEEE 802.11z) protocol. The TDLS wireless connection runs over a virtual tunnel, in the low level, it may be a direct connection between the source and sink or a virtual connection whose data is forwarded by the existing Wi-Fi AP. This mode is optional.

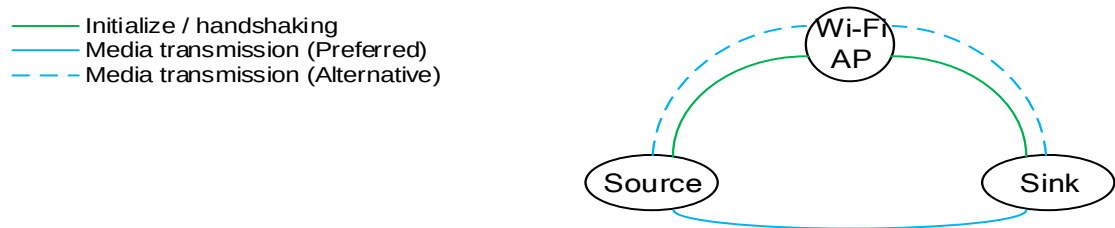


Figure 2. TDLS mode

3. Infrastructure mode

Defined in Miracast R2 specification. The source and sink are connected to an infrastructure network already, Miracast runs on the existing network directly, no need for additional network connection.

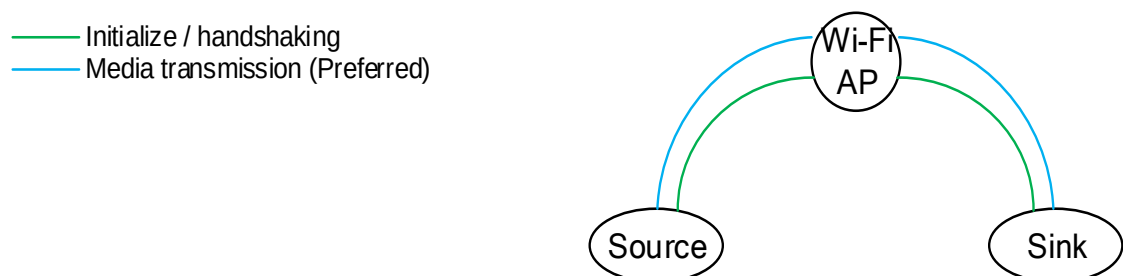


Figure3. TDLS mode over wireless infrastructure network

4. MS-MICE mode

Different from the previous 3 modes, this mode is not defined by Wi-Fi alliance officially. In fact, it is the abbreviation of Microsoft - Miracast over Infrastructure Connection Establishment Protocol. The handshaking runs still by Wi-Fi direct, but the subsequent media transmission can work on the infrastructure network which both source and sink are connected to. If there is no infrastructure network that both source and sink connected to together, the media transmission works on a connection built by Wi-Fi Direct (same as mode 1).

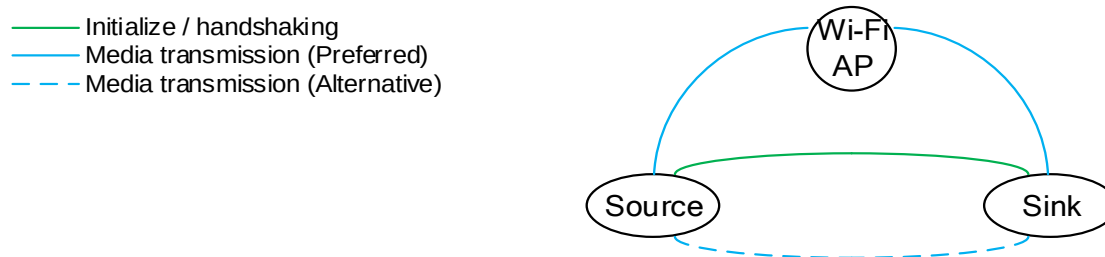


Figure4. MS-MICE mode with alternative wireless infrastructure network

As an extension defined by Microsoft, this mode is supported by Windows 10 1703 and newer version. Like mode 3, it can transmit media over infrastructure networks too. It is important to distinguish between the two.

2.2 Actual Applications

Four work modes are introduced in the previous chapter, their actual applications are not the same. The mode 1 is mandatory, so it is most common. The mode 4 is backed by Microsoft, it becomes more and more common now. Regarding mode 2 and 3, they are very rare. In fact, we never encounter a Miracast source or device which supports mode 2 or 3 and the support is already confirmed by the actual test.

Another concern involves concurrent Wi-Fi connections. Mode 1 is used widely, many users wonder below question: When the Miracast is running with Wi-Fi Direct mode, can the source device (mobile phone or laptop) connect to a Wi-Fi AP at the same time so to access the Internet or company network?

The answer is Yes! Most of the mobile phones and laptops produced in recent years can support concurrent Wi-Fi Direct and infrastructure connections perfectly.

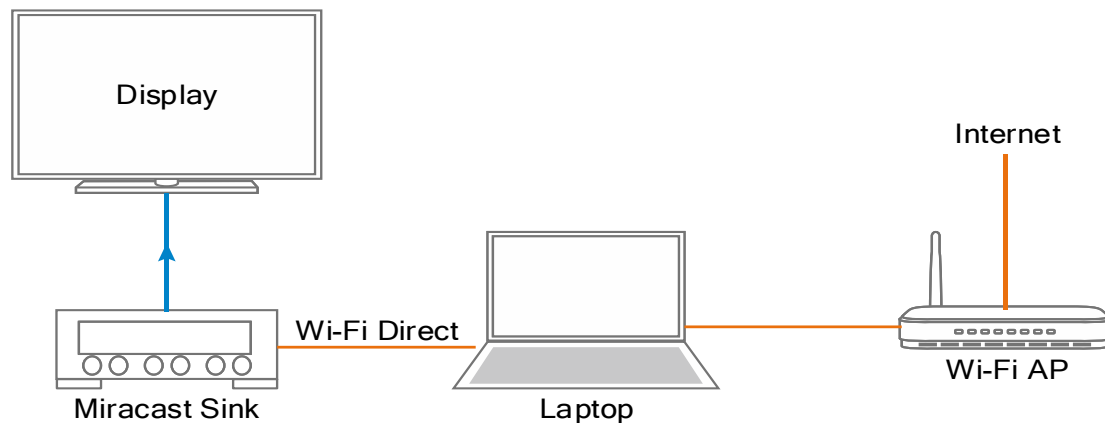


Figure5. Concurrent connections with Miracast and Wi-Fi AP

As shown in the above diagram, the laptop connects to a Wi-Fi AP so to access the Internet, at the same time, it projects the screen content to a Miracast sink over a Wi-Fi Direct connection.

3 Summary

The following table summarizes the characteristics of each of the four modes:

Mode	Initializing / Handshaking	Media Transmission	Actual Application
Wi-Fi Direct (P2P)	Wi-Fi Direct	Wi-Fi Direct	Most widely used
TDLS	Wireless infrastructure network	Virtual Direct connection over wireless infrastructure network	No actual application encountered
Infrastructure	Infrastructure network	Infrastructure network	No actual application encountered
MS-MICE	Wi-Fi Direct	Wi-Fi Direct or infrastructure network	More and more widely

Currently, our BYOD switcher supports Wi-Fi Direct (P2P) and MS-MICE modes well.

3.1 What Networking Modes are Supported?

Q

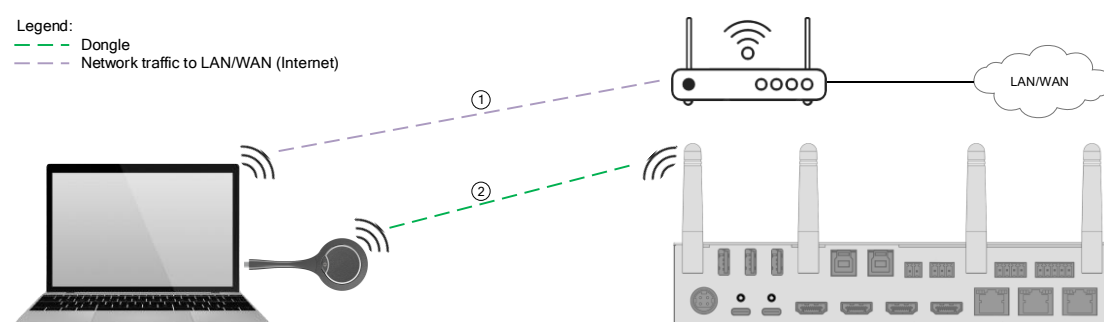
We are often asked questions related to networking, such as "Can this product

differentiate employees and visitors?" Please describe the supported networking modes.

A

The product has built-in Wi-Fi AP, 3 wired Ethernet interfaces and USB Ethernet bridge integrated in every USB-C input interface, furthermore, the firmware supports network isolation mode. All these designs provide powerful and flexible networking capability. Here are some common networking modes.

1. Simplest

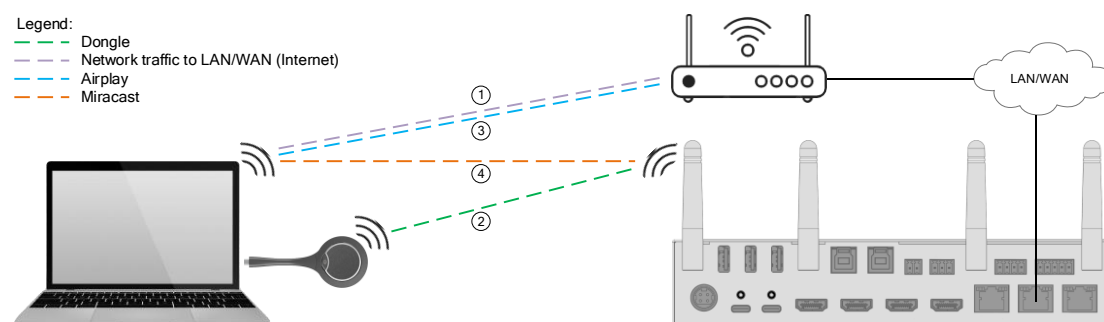


- ① Over the wireless connection to the external Wi-Fi AP, the computer can access the LAN/WAN (Internet).
- ② Over the wireless connection to the built-in Wi-Fi AP, the dongle can perform wireless casting and wireless conference.

For this networking mode, the computer can access the LAN/WAN wirelessly and use dongle to perform wireless casting and conference.

This networking mode is suitable for sites where dongle casting and conference is the main application scenario.

2. Basic



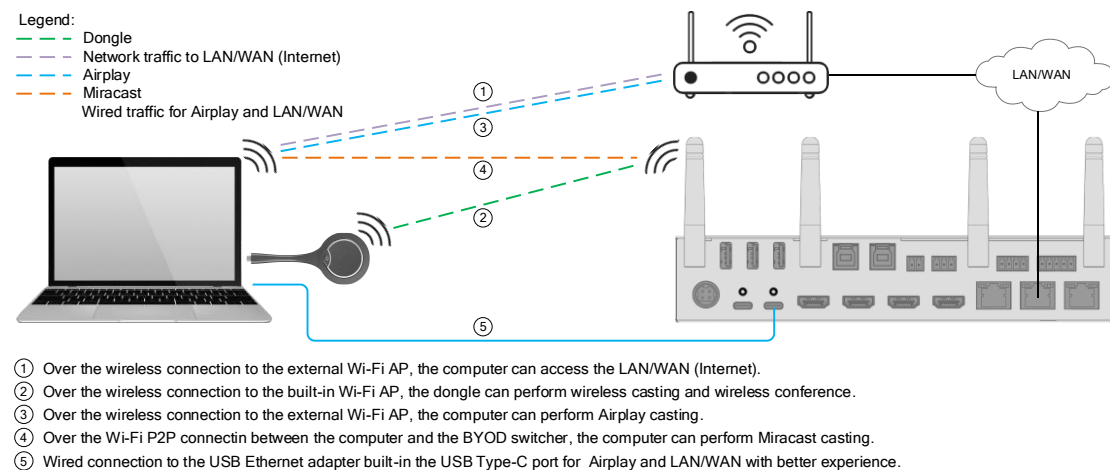
- ① Over the wireless connection to the external Wi-Fi AP, the computer can access the LAN/WAN (Internet).
- ② Over the wireless connection to the built-in Wi-Fi AP, the dongle can perform wireless casting and wireless conference.
- ③ Over the wireless connection to the external Wi-Fi AP, the computer can perform Airplay casting.
- ④ Over the Wi-Fi P2P connectin between the computer and the BYOD switch, the computer can perform Miracast casting.

Based on the simplest networking mode, this mode adds a wired connection between the BYOD switcher to the network of the site. For this networking mode, the BYOD switcher is connected to the network of the site, therefore the computer can perform Airplay casting over the wireless connection to the external Wi-Fi AP. Miracast is

supported too.

This networking mode supports all wireless casting modes, and it can be regarded as the basic networking mode suitable for most wireless occasions.

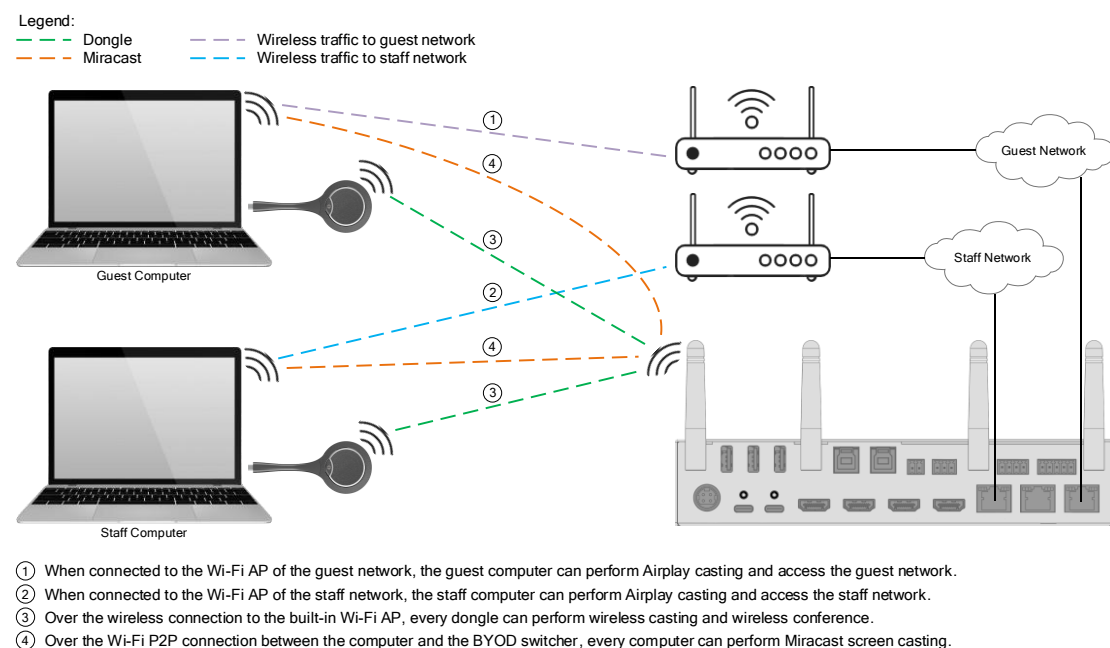
1. Recommendation



Based on the basic networking mode, this mode provides wired connection on the USB-C ports with built-in USB Ethernet adapter, so that the computer can perform Airplay casting and access the network with better experience.

It is recommended to use this networking mode whenever possible.

2. Secure

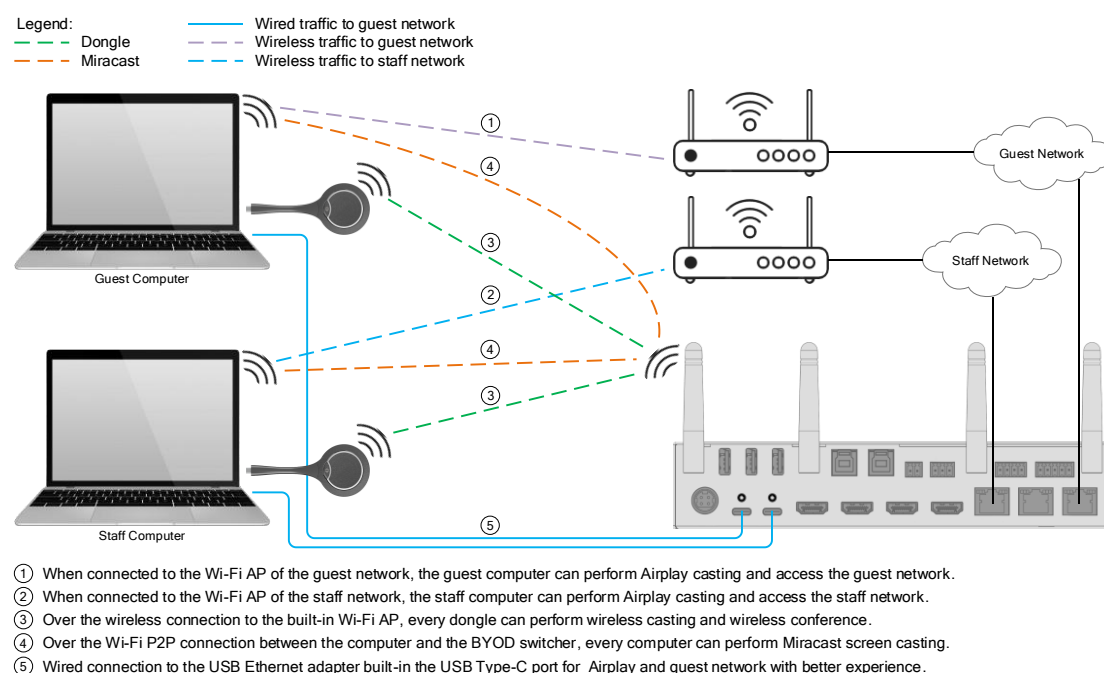


For this mode, the network isolation mode (Configuration "Secure Ethernet Mode" in Web UI) of the BYOD switcher is enabled, the **CONTROL** interface is connected to the

staff network and one of two **ETHERNET** interfaces is connected to the guest network. There is one Wi-Fi AP in each of the staff and guest networks. Airplay casting, Miracast, wireless casting and conference are supported for both guest and staff computers. Furthermore, the guest and staff computers can access resources on their respective networks.

This networking mode can limit wireless access users to access different network resources, it is suitable for the sites with higher security requirements.

1. Secure



Based on the previous networking mode, this mode provides wired connection on the USB-C ports, so that the computer can perform Airplay casting and access the network with better experience, for now all wired access users are limited to access the guest network only.

This networking mode can be used when needed.

3.2 How about External Wi-Fi AP?

Q

The BYOD switcher has built-in Wi-Fi module for wireless casting and conference, can I use an external Wi-Fi AP to extend the range or optimize the effect?

A

This question needs to be answered by distinguishing different screen casting protocol or methods.

1.Airplay

From the perspective of technology, Airplay runs over the infrastructure network, therefore the external Wi-Fi AP can be used for Airplay and the similar BYOD protocols (such as Google Cast). In the chapters of the previous question, several networking modes use external Wi-Fi AP for Airplay.

2.Dongle

The dongle and BYOD switcher have many optimizations for the characteristics of Wi-Fi communication so to provide sufficient and stable bandwidth for the wireless casting and conference applications, the effects in the same Wi-Fi environment are better than other vendors.

In view of the above, to ensure the effect of dongle, it is only recommended for the dongle to connect to the built-in Wi-Fi AP of the BYOD switcher directly.

3.Miracast

As the description in the previous chapter of this document, the two practical work modes of Miracast both depend on Wi-Fi direct (P2P) with which the connection must be built between the source and the sink directly. No relaying or forwarding is supported, so the external Wi-Fi AP cannot be used for Miracast.

There is a saying that the external Wi-Fi AP can extend the range, however, the BYOD switcher itself supports 30m of wireless casting and conference. This distance is enough for most application scenarios, so there is no need to use the external Wi-Fi AP to extend the range.